Comprehensive Design of Axisymmetric Wind Tunnel Contractions

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Design charts for wind tunnel contractions are developed using an inviscid, incompressible flow analysis. A contour formed of two cubic arcs joined smoothly together was found to be a good choice for a wall shape. Therefore, it was selected as the basis for a one-parameter family of wall shapes, which was then investigated in detail. The design chart parameters are the maximum wall pressure coefficients at the inlet (as an indicator of the danger of separation at the inlet end) and at the exit (which is related to the exit velocity nonuniformity). For any choice of these two parameters the charts yield the shape parameter and the nozzle length for this particular family of shapes. The charts may be used to design nozzles with no local separation at the inlet, and with any desired exit velocity uniformity. When the two pressure coefficients are chosen so that separation at both ends is just avoided, the exit boundary layer thickness should be near its minimum.

I Introduction

Practically all flow facilities and wind-tunnels are fitted with a contracting nozzle just ahead of the test section. The acceleration of the flow achieved in the nozzle serves several purposes, all of which are very important for the operation of a wind-tunnel. They are: (1) reduction of mean-flow nonuniformities to produce an even velocity profile at the test-section entrance, (2) reduction of the relative turbulence level, and (3) reduction of dynamic loads and losses in screens and honeycombs (due to reduced dynamic pressure in the settling chamber).

The most important parameter, by far, determining the magnitude of these effects is the area contraction ratio (CR). Once the value of CR is fixed, two other parameters take on importance, the contour and the nozzle length, which control the exit-velocity profile uniformity and the development of the boundary layer.

Exit-Velocity Uniformity. Intuitively it might appear that given a smooth transition into the straight section, a uniform flow in the test section may be obtained without any difficulties. It may be shown, however, that in all finite-length contractions, the wall-velocity does not increase monotonically but has a local minimum near the inlet and a local maximum near the exit. Let us consider the following argument. The governing equations are elliptic and, therefore, the contraction will affect the streamlines already in the straight section upstream, forcing them to curve. This streamline curvature produces a non uniform velocity profile within the straight section (Fig. 1)—with a defect at the wall near the wide end (and with an overshoot at the wall near the narrow end). Upon entering the contraction, the flow begins to accelerate *on the average*, but locally the curvature effects (transverse pressure gradients) still dominate. As a result, the wall velocity reaches a local minimum near the inlet of, but within, the contraction (Fig. 2). A similar argument indicates the existence of a local wall-velocity maximum near the exit of a finite contraction. As a result, the exit velocity profile is always nonuniform (Fig. 1). In short nozzles this nonuniformity may reach 10 percent or more, which is, of course, quite unacceptable in a wind-tunnel application. It is true that the flow becomes uniform downstream from the nozzle, but that still means that a part of the test section is lost for testing purposes.

Possibility of Separation. Due to the existence of wall-pressure extrema near the nozzle ends, there are two regions of adverse pressure gradient along the wall. The possibility of separation is of considerable concern because even localized separation causes thickening of the boundary layers, and large scale separation may produce flow unsteadiness. The pressure rise and the pressure gradients along the walls can certainly be alleviated by increasing the contraction length. However, this length is usually limited by available space and by cost. Also, the undesirable boundary layer growth in excessively long contractions has to be considered.

The problem of designing satisfactory wind-tunnel contractions has been the subject of many theoretical investigations [1–9].[1] However, in these works the tendency has usually been

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[1]Numbers in brackets designate References at end of paper.

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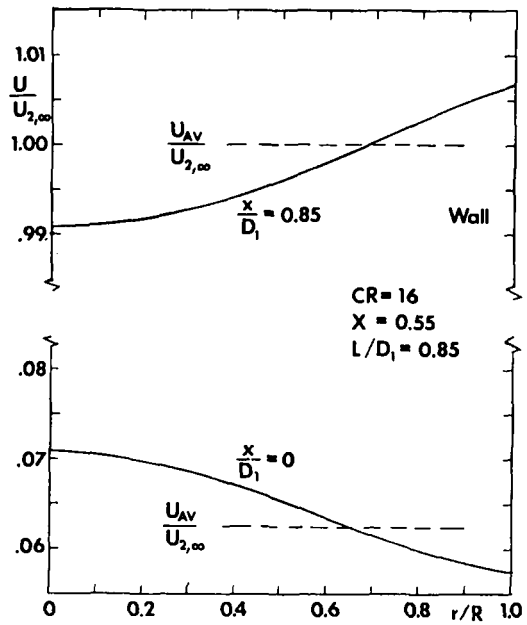


Fig. 1 Inlet and exit velocity profiles

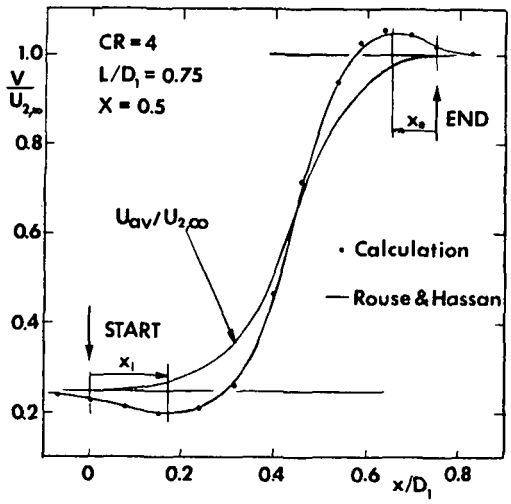


Fig. 2 Wall velocity distribution and the "one-dimensional" velocity distribution

to give more attention to the mathematical problem of solving the inviscid flow, rather than to the establishment of some practical criteria or guidelines that can actually be used by a designer.

The aim of the present work is to suggest what these criteria should be and to provide charts, based on inviscid flow calculations, in which the criteria may be used for design purposes.

II Previous Analytical Studies

In the absence of separation the flow within a contraction may be adequately described by the Laplace equation. This linear equation is relatively easy to solve for simple geometries (in contrast to the Navier-Stokes system) and, therefore, the problem of contraction design has attracted the attention of analytical investigators. As a result, many analytical solutions have been proposed for flows in both axisymmetric [1–5, 7, 9] and two-dimensional [5, 6, 8] contractions. The two-dimensional cases are usually solved in the hodograph (velocity) plane. A path is chosen in the hodograph plane describing the velocity along the wall of a contraction. The wall contour corresponding to this chosen velocity distribution is then obtained by transformation into the physical plane. The solutions for axisymmetric contractions are based on a series solution of the Laplace equation (for the velocity potential) or the Stokes-Beltrami equation (for the stream function) for a given velocity distribution along the centerline. The solution produces an infinite set of stream-surfaces and one searches for the most distant one (leading to the shortest contraction) with tolerable pressure gradients, to be used as the wall contour.

The fact that the closed-form (power series) solutions may be obtained for the contraction flow is certainly attractive. However, from the practical point of view, the above references offer little concrete design information about the optimum nozzle shape and length. In fact, it should be realized that these solutions are only mathematical tools to be used for the analysis of the flow. The application of these tools to contraction design requires that the designer first establish some *criteria* and then perform a trial-and-error search for the optimum shape and length. However, no suggestion for such a systematic search is given in any of the references. Last, but not least, is the problem that the solutions for axisymmetric nozzles are valid only for infinitely long nozzles (with exponentially decaying tails). Thus, the designer is also faced with the additional problem of truncating the nozzle at a "suitable location," and then he has to hope that the analytical solution is at least approximately valid for the truncated nozzle, too.

Because of drawbacks which these analytical solutions have, designers often use not-too-sophisticated methods of design such as (1) choosing a contour used previously elsewhere and believed to be "good," (2) applying some one-dimensional-flow techniques (but see Fig. 2 for illustration of the error involved), and (3) sketching a "plausible" shape. Such designs may

Nomenclature

a_1, a_2 = coefficients, see equation (13)
CR = contraction area ratio
$C_{pe} = 1 - (U_{2,\infty}/V_e)^2$
$C_{pi} = 1 - (V_i/U_{1,\infty})^2$
D = nozzle diameter
f = nondimensional wall shape function
F, G = functions defined by equations (7), (8), and (9)
$k = D/r_c$, dimensionless curvature
L = nozzle length
$m = D_1/D_2$
N = exponent
R_x = Reynolds number, see equation (11)
r_e = local longitudinal radius of curvature
s = distance defined in Fig. 7
U = velocity
$\tilde{u}_2 = (V - U_c)_2/U_{2,\infty}$
V = wall velocity
$X = x_m/L$, see Fig. 3
x = axial coordinate
x_0 = distance from the virtual origin of a boundary layer to the nozzle beginning
δ^* = boundary layer displacement thickness
$\xi = F_e^{1/3} G_i^{-1/2}$, see equation (10)
ρ = density

Subscripts

c = center line
e = point of wall-velocity maximum
i = point of wall-velocity minimum
1 = inlet plane
2 = exit plane
∞ = far upstream and downstream

Superscripts

(') = differentiation with respect to x
(•) = differentiation with respect to X/L

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actually prove to be satisfactory, but, generally, they are not optimum.

The only attempt, known to the author, to simplify the task of the designer by providing him with a simple-to-use design chart is the work of Rouse and Hassan [10]. They studied the behavior of the flow experimentally in an electrolytical bath, a procedure based on the electrical analogy to potential flows. The single chart they obtained is, however, intended as a criterion for avoidance of liquid cavitation near the exit, rather than for avoidance of separation. The possibility of separation at the low-velocity end appears not to have been considered by them at all and, therefore, their chart ought not to be used as the only criterion for the design of wind-tunnel nozzles.

III Present Approach

A broader study of the nozzle-design problem is attempted here, directed at the establishment of design charts allowing a comprehensive, yet rapid, design of an optimum axisymmetric nozzle for a given purpose.

This study is a numerical one which uses an approach converse to those described above; one starts with a given wall shape and solves for the velocities within it. If one desires to find the optimum contour and length, a trial-and-error procedure must be followed. In contrast to the previous analytical investigations, this approach does not yield closed form solutions, but it does solve the flow in true finite-length contractions. This was one of the reasons why the direct method of solution was used in this investigation. Another reason was the flexibility in wall contour selection severely restricted in the exact solutions.

In this analysis the decision was taken at the outset to investigate in detail only one suitable family of wall shapes, to reduce the number of independent variables. Further, attention was restricted, for the same reason, to one-parameter families only.

The type of one-parameter families considered always consisted of two matched curves, each having its apex at one end of the nozzle (Fig. 3). Six different curves were studied, all varying as a power of the axial coordinate, i.e., of the form x^N, with N equal 2, 2.5, 3, 3.5, 4, and 5. The comparison of their relative merits is postponed to the Discussion, because several concepts needed for the comparison are not introduced in the paper before then. In summary, the cubic was judged to be a good choice for a simple one-parameter family of wall shapes and was, therefore, chosen for a deeper analysis. (The cubics were also chosen by Rouse and Hassan [10] in their study). All the results presented here pertain to this family, Fig. 3, with $X = x_m/L$ as the sole representative of the contour shape.

All calculations were made using a computer program of the streamline curvature type, developed by General Motors Detroit Diesel-Allison Division as an extension of the original computer program of Katsanis [11]. The program is a finite-difference procedure solving Euler's equations of motion (compressible) in the intrinsic coordinates. The boundary conditions are the nozzle geometry, inlet profiles of stagnation pressure and temperature (taken here to be uniform) and the total mass flow. The maximum Mach number reached here was $M_2 = 0.07$ (for CR = 25), i.e., the flow was incompressible. The calculation starts with an assumed initial distribution of streamlines on which it iterates until all equations and boundary conditions are satisfied. In the present calculations all contractions were preceded and followed by pieces of straight duct 0.7 local diameter long. No further lengthening has any effect on the results, as the velocity profile nonuniformity at these end points is below one percent of the nonuniformity in the inlet and exit planes, respectively.

The assumption underlying the calculations presented below is that the flow is, in the absence of separation, adequately described as potential flow. As is usual in the boundary layer theory, one can assume that the presence of the boundary layer will not change the potential-flow pressure distribution as long as the boundary layer is thin. In the present context it requires $\delta^*/R << 1$, a condition which is well satisfied in most wind tunnels.

IV Design Criteria

In order to fully define this engineering problem, one has to formulate the design criteria, which were already touched upon in the Introduction. In summary, once the contraction ratio is chosen, the design criteria in wind-tunnel contraction design are: exit flow uniformity, separation, exit boundary layer thickness, and space/cost. The design parameters are the length, wall shape, and Reynolds number.

Of these four criteria the most important are the first two. Therefore, the design procedure proposed here is directed mainly to avoidance of separation and to producing the exit velocity profile with a prescribed nonuniformity defined as

$$\tilde{u}_2 \equiv (V - U_c)_2/U_{2,\infty}$$

where V is the wall velocity, U_c is centerline velocity, and subscript 2 refers to the conditions in the exit plane of the nozzle. $U_{2,\infty}$ is the velocity far downstream.

The present numerical investigation has shown that, for the cubic family of wall shapes, the only information the designer needs to know about the flow are the values of two wall-pressure coefficients defined as

$$C_{pe} \equiv 1 - (U_{2,\infty}/V_e)^2$$

$$C_{pi} \equiv 1 - (V_i/U_{1,\infty})^2$$

where subscripts e and i refer to the point of maximum and minimum wall velocity, respectively (Fig. 2). This very important simplification is due to the fact that the wall pressure distributions are quite similar. As a result $dC_{pi}/d(x/D_1)$ and $dC_{pe}/d(x/D_2)$ are mainly a function of C_{pi} and C_{pe}, respectively, and only a mild function of the nozzle geometry. Similarly, $\tilde{u}_2$ is found here to be proportional to C_{pe} (at least for $C_{pe} < 0.2$) and depends only very weakly on the geometry. Consequently, the results of this investigation are presented in the design charts only in terms of these two C_p's.

The design problem is thus reduced to a priori determination (or choice) of C_{pi} and C_{pe}. The design charts are then used to produce the corresponding nozzle length and shape. The choice of C_{pi} may be done using the Stratford's separation criterion along the lines suggested in the Discussion. As for $\tilde{u}_2$, it is left up to the designer to set the maximum value he can tolerate, taking into account the rapid decay of the nonuniformity downstream (see Section V). The choice of $\tilde{u}_2$ is always a compromise, and it involves a trade-off between $\tilde{u}_2$ and the nozzle length (see Discussion).

Too long contractions generate relatively thick exit boundary layers due to their length. Too short contractions have the same problem due to separation. One may expect that the exit boundary layer thickness has a minimum at some intermediate length.

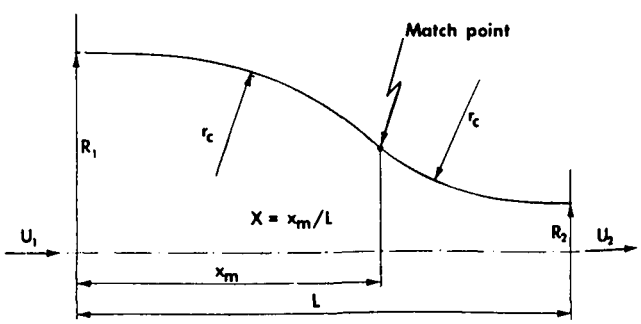


Fig. 3 Wall contour constructed of two matched cubic arcs

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It is conjectured here that this minimum will be approached by a design with maximum tolerable C_{pi} and C_{pe} ($\tilde{u}_2$). Such design tends to satisfy all four of the design criteria to a reasonable degree, and provides a desirable exit flow within the shortest nozzle length.

V Numerical Results

The calculations were performed at discrete values of the three geometrical parameters: CR = 2, 4, 9, 16, 25, L/D_1 = 0.75, 0.85, 1.0, 1.25, and X = 0.2, 0.3, 0.425, 0.575, 0.7, 0.8 (plus 0.12 for CR = 2), i.e., 124 configurations altogether.

The results obtained for three of the contraction ratios, CR = 4, 9, and 16, spanning the most common range of interest, are presented in Figs. 4, 5, and 6. (The results for all five contraction ratios are, however, presented together in the "unified" plots of Section VI.) An inspection of Figs. 4, 5, and 6 reveals the influence of the parameters X, L/D_1, and CR on the C_p's. It is seen that by changing X one can trade an increase in C_{pe} for a decrease in C_{pi}, or vice versa. The difference between the values of C_p's at L/D_1 = 0.75 and 1.25 is very large; this range of L/D_1 covers most of the nozzles that would be used for wind-tunnel purposes. The most important effect of the contraction ratio is the shift of the C_p curves toward larger X as CR increases.

The application of the Stratford's separation criterion (see Discussion) requires the knowledge of the pressure gradients and of the location of the pressure extrema (i.e., distances x_i and x_e of Fig. 2). The pressure gradient near the inlet was approximated by

$$\frac{dC_p}{d(x/D_1)} = 0.4\,\frac{C_{pi}}{s/D_1}$$

where s is the distance between the points where $C_p = 0.4\,C_{pi}$ and $0.8\,C_{pi}$. The results, presented in Fig. 7, show a good collapse of the data especially for CR $\geq$ 9. The pressure gradient is seen to be primarily a function of C_p and to depend only mildly on the details of geometry. Similarly, on the exit side the pressure gradient $0.4\,C_{pe}\,D_2/s$ is equal about $4\,C_{pe}$ (for $C_{pe} \leq 0.1$).

The positions of C_{pi} and C_{pe}, x_i, and x_e, are plotted in Figs. 8 and 9 against the distance from the nozzle ends to the match points of cubics. The data collapsed satisfactorily and exposed two different asymptotic behaviors. For small $XL/D_1[(1 - X)L/D_2]$, associated with high wall curvatures, $x_i/D_1(x_e/D_2)$ increases linearly. At high values of the abscissa, however, a limit is reached with x_i/D_1 and x_e/D_2 equal about 0.25.

Another result obtained from the calculations was the dependence of $\tilde{u}_2$ on C_{pe}. It was found that a simple linear relation holds between the two, with only a very mild dependence on the geometry. For all cases where $C_{pe} < 0.2$, $\tilde{u}_2 = (0.35 \div 0.4)\,C_{pe}$. This nonuniformity, caused by the streamline curvature alone, decays rapidly within the test section. It reduces to 0.45, 0.21, 0.1, 0.049, 0.025 of $\tilde{u}_2$ in the distance x/D_1 = 0.1, 0.2, 0.3,

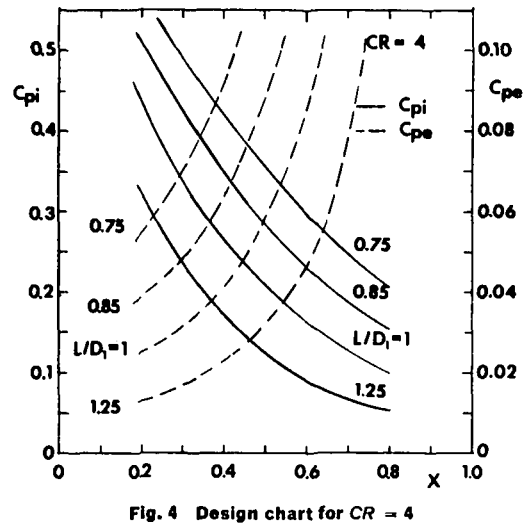


Fig. 4 Design chart for $CR = 4$

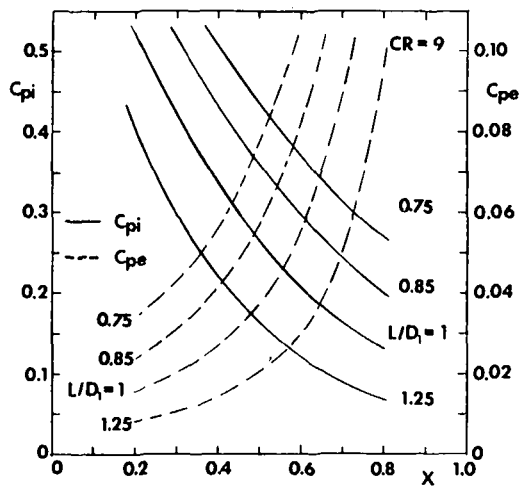


Fig. 5 Design chart for $CR = 9$

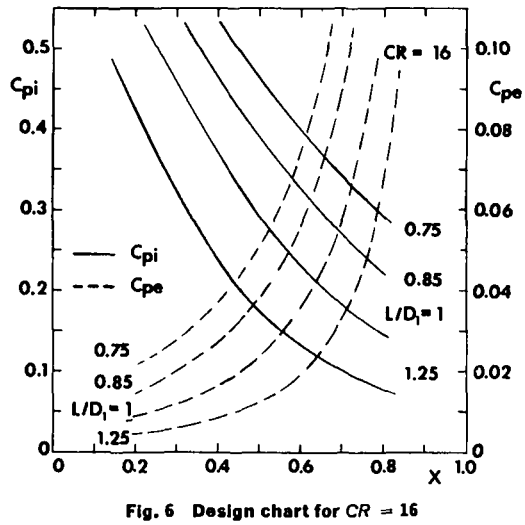


Fig. 6 Design chart for $CR = 16$

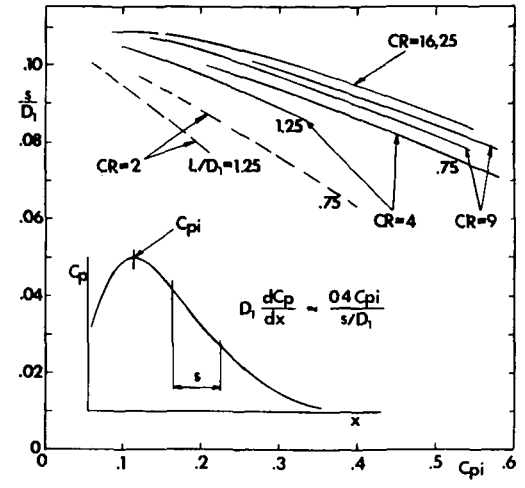


Fig. 7 Wall-pressure gradient

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0.4, 0.5, respectively. This decay may be approximated in the above range by exp $(-7.5\ x/D_1)$. (It may be readily shown, by solving the Laplace equation, that sufficiently far downstream from the exit plane the nonuniformity must decay as exp $(-7.66\ x/D_1)$).

Finally, it was found, by cross-plotting the data of Figs. 4, 5, and 6, that when C_{pi} and C_{pe} are held fixed, the required L/D_1 *decreases* with increasing contraction ratio, contrary to the intuition. This surprising result is explored further in the next section.

It should be noted that one also has to consider the Reynolds number in addition to the three geometrical parameters of the problem CR, L/D_1, and X (which are the only parameters that appear in the inviscid formulation). This important parameter is not neglected; it enters the procedure at the point where the designer makes his choice of appropriate maximum allowable C_p's.

A comparison may be made of the present calculations with one example obtained using the electrical analogy approach in reference [10]. As seen in Fig. 2, the agreement of the two methods is good.

VI Scaling Parameters for C_{pi} and C_{pe} Obtained From Simplified Analysis

The rather surprising finding that nozzles with large CR require smaller L/D_1 for the same C_p's than nozzles with small CR merits some attention. Intuition would seem to suggest that the length should increase, perhaps proportionally, with the diameter difference

$$L \sim (D_1 - D_2) \tag{1}$$

However, this expectation is not borne out by the present results. The apparent discrepancy prompted an attempt to reconcile it using a simple analysis. Another reason for the analysis was the need of some qualitative information about suitable scaling parameters for the wall-pressure coefficients C_{pi} and C_{pe}, allowing a more general presentation of the data.

As already noted, the existence of the regions of adverse pressure gradient is caused by the wall curvature. Consequently, it is plausible to assume that the C_p's are related to some characteristic value of the wall curvature. Let us consider wall-curvature rendered nondimensional by the local nozzle diameter, D:

$$k = \frac{D}{r_c} = \frac{1}{2}\frac{DD''}{[1 + (D'/2)^2]^{3/2}} \tag{2}$$

where prime denotes differentiation with respect to the axial coordinate x. Further, let us consider a family of nozzles such that

$$\frac{D - D_2}{D_1 - D_2} = f(x/L;\ p_i) \tag{3}$$

where p_i is a parameter array. Near the nozzle ends, where $(D')^2 << 1$ and $D \approx D_1$ or D_2, one can write

$$k_{\text{inlet}} \approx \frac{1}{2}\frac{m-1}{m}\left(\frac{D_1}{L}\right)^2 \ddot{f} \tag{4a}$$

$$k_{\text{exit}} \approx \frac{1}{2}\frac{m-1}{m^2}\left(\frac{D_1}{L}\right)^2 \ddot{f} \tag{4b}$$

where dot indicates differentiation with respect to x/L. To keep C_p's small, one has to keep the wall curvature near the nozzle ends small as well. In addition, k should grow gradually at the nozzle juncture with straight sections, in order to avoid a step change in transverse and longitudinal pressure gradient at the juncture. It follows that one should use nozzle shapes with a fairly rapid change of area in the central portion, but a gradual change at the ends, with $\ddot{f} = 0$ at the end points.

In the case of matched cubics

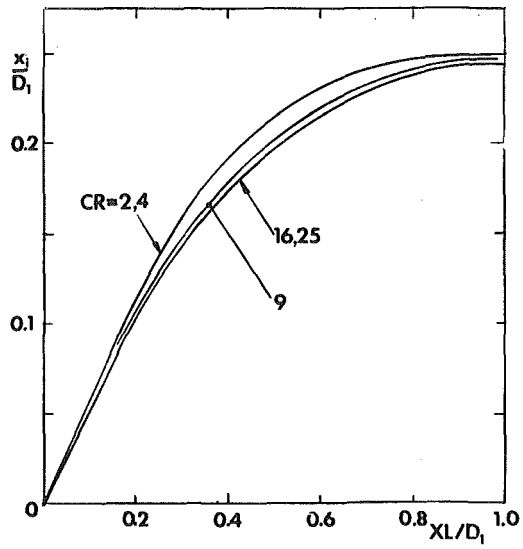


Fig. 8 The distance from the nozzle beginning to the wall pressure maximum versus the distance from the nozzle beginning to the cubic match-point

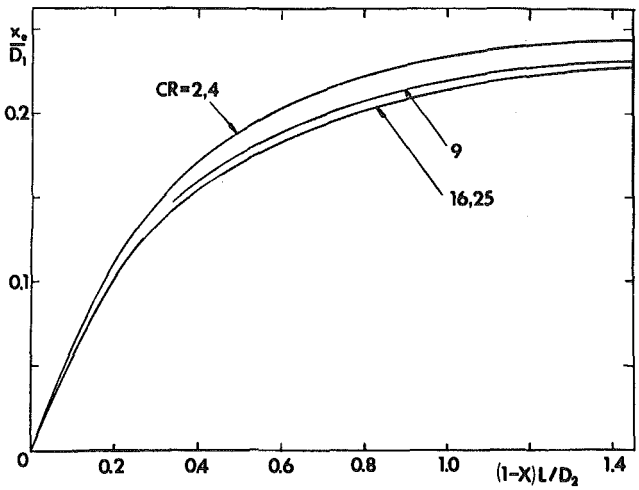


Fig. 9 The distance from the nozzle end to the wall pressure minimum versus the distance from the nozzle end to the cubic match-point

$$f = 1 - \frac{1}{X^2}\frac{x^3}{L} \qquad x/L \le X \tag{5a}$$

$$f = \frac{1}{(1-X)^2}\left(1 - \frac{x^3}{L}\right) \qquad x/L > X \tag{5b}$$

which gives near the ends

$$k_{\text{inlet}} \approx -3\frac{m-1}{m}X^{-2}\left(\frac{L}{D_1}\right)^{-2}\frac{x}{L} \tag{6a}$$

$$k_{\text{exit}} \approx 3\frac{m-1}{m^2}(1-X)^{-2}\left(\frac{L}{D_1}\right)^{-2}\left(1 - \frac{x}{L}\right) \tag{6b}$$

It may be ssen that near the ends the curvature varies linearly and is equal zero at the nozzle ends. It may be shown that away from the ends this trend changes, however, and k reaches a local extremum.

To proceed further, one has to make some assumptions tying the C_p's to some characteristic value of the wall curvature. There are two possible choices:

(*a*) Make the characteristic value of k equal the *maximum* value of the cubic-curve curvature, which may be shown (for

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the inlet side) to reach its maximum at $x/D_1 = 0.546\ F_i^{-1/2}$ with the value of

$$k_{\text{inlet max}} = 1.246\ F_i^{1/2} \tag{7a}$$

where F_i is defined by the wall contour equation

$$\frac{D}{D_1} = 1 - F_i \left(\frac{x}{D_1} \right)^3 \tag{7b}$$

i.e., from equations (3) and (5a)

$$F_i = \frac{m-1}{m} X^{-2} \left(\frac{L}{D_1} \right)^{-3} \tag{7c}$$

Similarly for the exit side

$$|k_{\text{exit}}|_{\max} = 1.246\ F_e^{1/2} \tag{8a}$$

and

$$F_e = \frac{m-1}{m^3} (1 - X)^{-2} (L/D_1)^{-3} \tag{8b}$$

(*b*) Make it equal the *local* curvature at x_i (or x_e). Assuming that

$$\frac{x_e/L}{X}, \frac{1 - x_i/L}{1 - X} \approx \text{constant} \tag{9a}$$

and substituting this expression into equation (6) one can write

$$k_{\text{inlet max}} \sim \frac{m-1}{m} X^{-1}(L/D_1)^{-2} = G_i \tag{9b}$$

$$k_{\text{exit max}} \sim \frac{m-1}{m^2} (1 - X)^{-1}(L/D_1)^{-2} = G_e \tag{9c}$$

The above reasoning produced four dimensionless groups F_i, F_e, G_i, G_e which one may attempt to use to collapse all the C_p data obtained from the numerical studies. The purpose of such an attempt is to provide a more concise form of data presentation. If the data really do collapse, then the tedious task of interpolating for intermediate values of CR and L/D_1 from the design charts, Figs. 4, 5, and 6, is eliminated. Also, an important advantage is that the trends in variation of the parameters are then clearly visible.

All data were actually plotted against the four groups (two for inlet, two for exit). The data were seen to collapse reasonably well, depending on the values of C_p's. For $C_p < 0.1$, a collapse was achieved when plotting the C_p's against F_i and F_e (Fig. 10). For $C_{pi} > 0.1$ it was G_i which allowed a very good collapse for any constant CR (Fig. 11). However, some variation with CR persisted, which could conceivably be remedied by inserting an extra m-dependence into the definition of G_i. Fortunately, the variation with CR is small enough to allow an adequate interpolation in the graph and so no modification is really necessary . The plot of $C_{pe} > 0.1$ against G_e is not presented as it has little practical value for wind tunnel design.

The change of functional relationship from $C_p = C_p(F)$ to $C_p = C_p(G)$ around $C_p = 0.1$ is apparently related to the behavior of $x_i/D_1(x_e/D_2)$ with respect to XL/D_1 ($(1 - X)L/D_1$) shown in Figs. 8 and 9. Near the origin (x/D)'s increase linearly in agreement with assumption (9a), and C_p's scale with G's. Further away (x/D)'s become independent of X in agreement with the assumption that the other (matching) cubic is far enough for its effect to be felt, and C_p's scale with F's.

The information contained in Figs. 10 and 11 may be used to set up a simple procedure for contraction design, under the following restrictions

$$C_{pi} > 0.1 \qquad 0.75 \le L/D_1 \le 1.25$$

$$C_{pe} < 0.1 \qquad 0.2 \le X \le 0.8$$

$$2 \le \text{CR} \le 25$$

Then, for any chosen CR, C_{pi}, and C_{pe} one obtains from Figs. 10 and 11 the appropriate values of F_e and G_i. Manipulation of equations (8b) and (9b) leads to an implicit equation for X

$$X^{1/2}(1 - X)^{-2/3} = F_e^{1/3} G_i^{-1/2} m^{1/2}(m - 1)^{1/6} = \xi m^{1/2}(m - 1)^{1/6} \tag{10}$$

which yields the appropriate X for any given ξ and m. L/D_1 is then obtained from equation (9b). To facilitate the evaluation of equation (10), especially the extraction of X from the expression on the left-hand side, one can use the graphical solution of Fig. 12. The set of equations (9b) and (10) allows the study of the behavior of L/D_1 with CR for a fixed ratio ξ, i.e., approximately fixed C_{pi} and C_{pe}. This behavior is explored in Fig. 13 in terms of the ratio of L/D_1 to $(L/D_1)_\infty$—the limiting case for CR $\to \infty$. The lines of constant ξ cover the range of normal interest (e.g., for $C_{pi} = 0.35$ and $C_{pe} = 0.04$, $\xi \approx 0.51$). It may be observed that the required L/D_1 actually decreases with increasing contraction ratio for CR greater than about four. This trend becomes more pronounced as ξ decreases. For small CR the trend is reversed, however, so that for $R_2/R_1 \to 1$, $L/D_1 \to 0$.

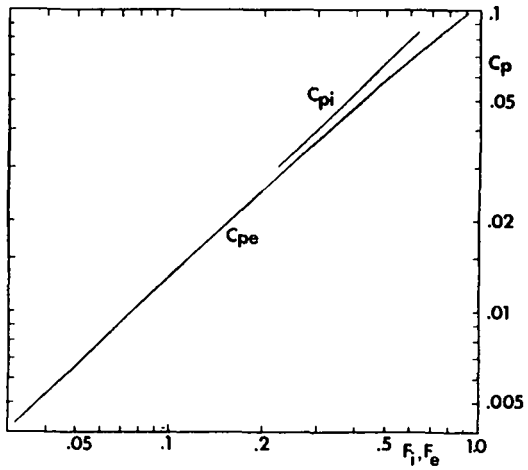


Fig. 10 Dependence of C_{pi} and C_{pe} on the dimensionless parameters F_i and F_e respectively

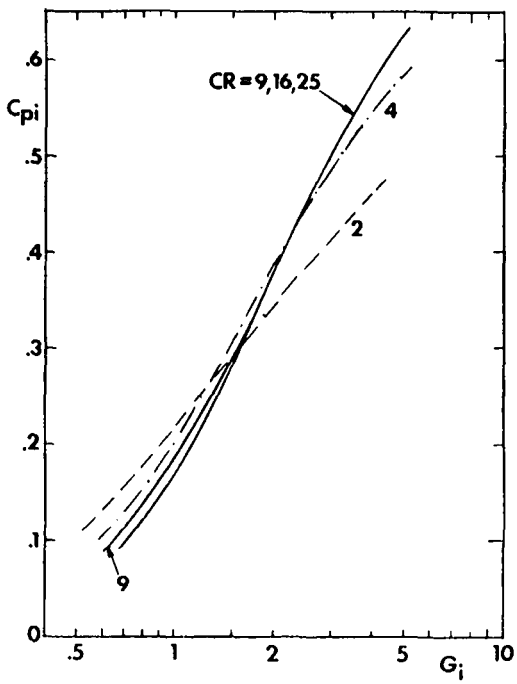


Fig. 11 Dependence of C_{pi} on the dimensionless parameter G_i

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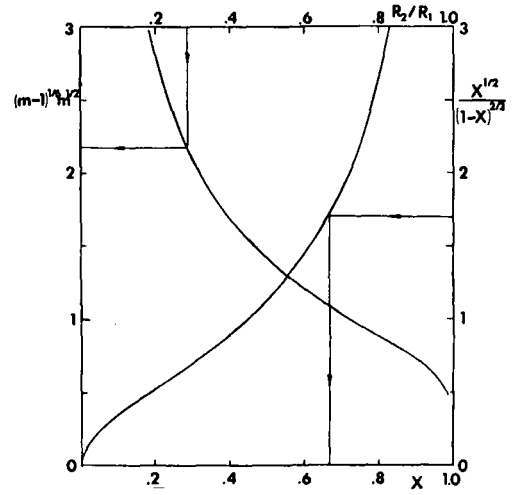


Fig. 12 Graphical aid for evaluation of X from equation (10)

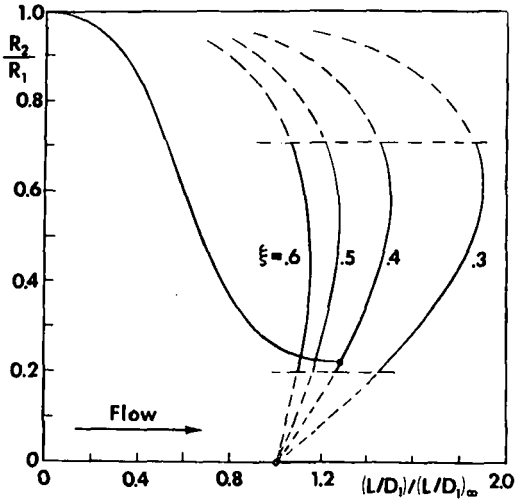


Fig. 13 Dependence of L/D_1 on R_2/R_1 at fixed values of $\xi = F_s^{1/3} G_i^{-1/3}$. The extrapolated portions of the curves are indicated by broken lines

VII Discussion

The purpose of this chapter is to bring up several subjects concerning the application of the presented design charts.

Separation Criterion at the Wide End of a Nozzle. Wall pressures in contractions with a small L/D_1, or with a small X, may reach values of $C_{pi} > 0.5$ and that means there is a real possibility of separation. In most wind-tunnels the boundary layer will become turbulent (due to high enough Reynolds number and the adverse pressure gradient) before the region of large dC_p/dx and C_p is reached. Whether or not the turbulent boundary layer will separate may be investigated using the Stratford's separation criterion [12], stating that the separation will occur when

$$C_p \left(x \frac{dC_p}{dx} \right)^{1/2} = 0.35 \, (10^{-6} R_x)^{0.1} \qquad (11)$$

where

$$C_p = 1 - \left(\frac{U}{U_m} \right)^2$$

$$R_x = U_m x/\nu = 0(10^6)$$

$$d^2p/dx^2 \leq 0$$

and where U_m is the velocity at the beginning of the pressure rise and x is the local distance from the virtual origin of the turbulent boundary layer.

A simple working form of this criterion may be obtained by using several results of Section V. To this end let us be concerned only with nozzles where separation just sets in, in order to find the upper bound on C_{pi} below which separation does not occur. For such nozzles one can make the following approximations:

(*a*) Separation will occur close to the point of maximum pressure, say at 0.95 C_{pi}.

(*b*) The pressure gradient may be approximated by the average pressure gradient between the points $C_p = 0.4\ C_{pi}$ and $0.8\ C_{pi}$ (Fig. 7), i.e.,

$$dC_p/d_x \approx 0.4\ C_{pi}/s$$

(*c*) The distance x consists of two parts: the distance from the beginning of the nozzle to the separation point, and x_0, the distance from the virtual origin of the boundary layer to the beginning of the nozzle. The former length is always somewhat less than x_i (presented in Fig. 8), say $0.9\ x_i$. The latter length may be taken as the distance to the nearest turbulence screen upstream.

Consequently, one may write

$$C_p \left(x \frac{dC_p}{dx} \right)^{1/2} = 0.6\ C_{pi}^{\ 3/2} \left(\frac{x_0 + 0.9\ x_i}{s} \right)^{1/2}$$

Substituting into equation (11) yields

$$C_{pi} = 0.7 \left(\frac{x_0 + 0.9\ x_i}{s} \right)^{-1/3} (10^{-6} R_x)^{1/15} \qquad (12)$$

This equation may be used as a guide in selection of C_{pi}. In a typical example, with $x_0/D_1 = 0.3$, $x_i/D_1 = 0.15$, $s/D_1 = 0.09$, and the settling chamber Reynolds number based on D_1 equal 10^6, equation (12) yields

$$C_{pi} = 0.7 \left(\frac{0.435}{0.09} \right) 0.435^{1/15} = 0.39$$

Design Criteria at the Narrow End of a Nozzle. There are two criteria which have to be satisfied at the narrow end of a nozzle: the maximum acceptable flow nonuniformity and the avoidance of separation. In most practical cases $\tilde{u}_2$, the nonuniformity, will be required to be no more than 2 percent. For cubic nozzles, where $\tilde{u}_2 \approx 0.35\ C_{pe}$, this translates into, say, $C_{pe} \leq 0.06$. Since, from Section V, $dC_p/d(x/D_2) \approx 4\ C_{pe} \leq 0.24$ (i.e., quite small), there is no danger of separation for a turbulent boundary layer. The laminar boundary layer, however, can overcome only a much smaller pressure rise than a turbulent one and some conconsideration must be given to separation. Fortunately, the pressure distribution around the point of minimum pressure is such that the boundary layer will tend to resist separation. First, the boundary layer is well energized at the pressure minimum after the region of favorable pressure gradient and can better resist the pressure rise that follows. Second, the pressure recovery is faster at the beginning than toward the end of the deceleration, which favors a greater pressure recovery. Due to this, the design constraint at the narrow end is usually only the maximum allowable $\tilde{u}_2$.

Total Nozzle Length. As already noted, the flow is neither parallel nor uniform for some distance within the straight sections ahead of and after the contraction. These distances may be viewed as de facto parts of the nozzle. Therefore, for practical purposes, one may introduce the concept of the "total nozzle length"

$$L_t = L + a_1 D_1 + a_2 D_2 \qquad (13)$$

where $a_1 D_1$ and $a_2 D_2$ are the distances upstream and downstream within which local velocity-profile distortions are still

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important. The distance a_1D_1 is the closest distance upstream from the start of a nozzle where one may locate the last turbulence-reducing screen without causing a flow distortion and a nonuniform pressure drop across that screen. The distance a_2D_2 is the closest distance downstream from the nozzle end where the effects of nozzle curvature decay to negligible levels—the effective starting point of the test section. (For example: in a nozzle where $C_{pi} = 0.35$, $a_1 \geq 0.2$ to achieve 3 percent velocity nonuniformity at $x = -a_1D_1$, $a_2 = 0.3$ is sufficient to reduce $\tilde{u}_2$ to one tenth, whatever the initial value of u_2, at $x = L + a_2D_2$). Shortening of the nozzle length L by some ΔL always results in an *increase* in the other two lengths, and so the saving in the total length L_t is always less than ΔL. This effect grows rapidly with decreasing L/D_1—a fact which ought to be considered in nozzle design.

Comparison of Different Power-Law Wall Shapes. As already mentioned in Section III, six different power-law curves were tested, with orders $N = 2$, 2.5, 3, 3.5, 4, and 5. One particular set of geometrical parameters was chosen for the comparison, with CR = 9, $L/D_1 = 1.0$ and $X = 0.5$ (Table 1). It is seen that increasing N causes a monotonic increase of C_{pi}, while C_{pe} has a minimum at about $N = 4$; $\tilde{u}_2$ decreases monotonically. Of interest is the last column containing the pressure gradient term needed in equation (12). This term is almost unaffected by N, and so for separation purposes the families may be compared on the basis of C_{pi} alone. For this reason, the second line under each family gives the results for "equivalent" cubic nozzles with the same CR, L/D_1, and C_{pi} (but different X shown in parentheses).

Even when the different wall shapes are compared in this manner, the exit plane nonuniformity is still seen to decrease monotonically with N. (Downstream from the exit plane the nonuniformity decays according to the same exponential law for all orders.) However, since the ratio $C_{pe}/\tilde{u}_2$ increases rapidly with N, the major design criterion at the exit end tends to become C_{pe} (separation) rather than $\tilde{u}_2$, depending on how small the prescribed $\tilde{u}_2$ is. On the other hand, the cubic is the shape which produces the lowest combination of C_{pi} and C_{pe} of all the curves. This means that the cubic is most suitable for designs calling for the shortest nonseparated nozzle. For this reason it was chosen for this study.

Concluding Remarks. It may be observed that, due to their large acceleration, nozzles with large contraction ratios are tolerant to any irregularities at the low velocity end, such as separation-induced thick boundary layers and unsteadiness. On the other hand, nozzles with smaller CR (say 4 or 5) demand that greater attention be given to the avoidance of separation.

The application of the present results to rectangular contractions must be done with some caution. The main problem is the existence of the circumferential pressure gradients. Fortunately, the favorable streamwise pressure gradient acts to reduce the boundary layer thickness, thus alleviating the problem, especially in the case of a large contraction ratio.

Table 1 Comparison of different power-law wall shapes

N (X)	C_{pi}	C_{pe}	$\tilde{u}_2$	$\frac{C_{pe}}{\tilde{u}_2}$	$\frac{x_i}{D_1}$	s	$\left(\frac{x_i+.3}{s}\right)^{-1/3}$
2	.217	.074	.0510	1.45	.07	.062	.55
(.58)	.217	.052	.0195				
2.5	.228	.048	.0244	1.97	.14	.08	.57
(.54)	.228	.044	.0165				
3	.267	.037	.0129	2.86	.20	.098	.58
3.5	.32	.033	.0074	4.4	.25	.105	.575
(.42)	.32	.028	.0105				
4	.37	.032	.0046	7.0	.29	.110	.57
(.36)	.37	.024	.0080				
5	.473	.036	.0020	18.2	.33	.115	.565
(.25)	.473	.018	.0067				

Conclusions

1 The one-parameter family of wall shapes formed of two smoothly joined cubics (with the match point location X as the parameter) is a suitable choice for a wall contour. It is very efficient—it achieves the prescribed C_p's within a relatively short nozzle length L.

2 All the information necessary to design a nozzle of the given family is contained in the values of C_{pi} and C_{pe}, the maximum wall pressure coefficients (for CR $\geq$ 4). This is so because the pressure gradients are very nearly functions of the C_p's alone. Also, the exit nonuniformity $\tilde{u}_2$ is found to be related uniquely to C_{pe}.

3 It is important to realize that there is not one optimum wall shape suitable for all contractions, not even for a given contraction ratio. To find a good shape, one has to take into consideration the ultimate application of the nozzle to arrive at the maximum acceptable levels of C_{pi} and C_{pe}, which ought to be the starting point of a comprehensive design.

4 In the case of cubic wall shapes (but not necessarily for any other shapes), the exit nonuniformity $\tilde{u}_2$ is about 0.7 to 0.8 times the maximum wall-velocity overshoot $\tilde{u}_e$ ($\approx 1/2\ C_{pe}$) which occurs upstream within the nozzle at the distance of about 0.2–0.25 test-section diameters from the nozzle end.

5 The only criterion for choosing C_{pi} is the avoidance of separation near the inlet, and here the value given by equation (12) is probably a good upper bound below which a turbulent boundary layer will not separate. The criterion on C_{pe} will usually be the desired exit-velocity uniformity ($C_{pe} = 2.5 - 3\tilde{u}_2$). Once C_{pi} and C_{pe} are chosen, X and L/D_1 can be obtained from the design charts. Note that neither C_{pi} nor C_{pe} can be reduced to zero in a finite-length contraction.

6 C_{pi} and C_{pe} change so rapidly with L/D_1 that the relatively narrow range $0.75 \leq L/D_1 \leq 1.25$ covers most of the wind-tunnel nozzles of practical interest. The nozzle is often preceded by a straight section, the settling chamber, which contains screens and honeycombs. It is advisable to locate the last screen at least 0.2 D_1 upstream of the nozzle, to avoid significant distortion of the flow through the last screen. On the high-velocity end, the exit velocity-profile nonuniformity is reduced to about 10 percent within 0.3 D_2 of the following straight section.

7 The present results suggest that one really should introduce the concept of a "total nozzle length" encompassing the entire region where the flow is significantly nonuniform and nonparallel (which includes parts of the straight sections ahead of and after the nozzle). The underlying reason for it is that shortening of the nozzle itself reduces only the curved part of the "total length," while it lengthens the straight portions. This effect is especially strong in short nozzles.

8 For a given choice of C_{pi} and C_{pe}, in the CR range investigated, L scales with D_1 rather than the ΔD. It is found here that for large contraction ratios, the needed L/D_1 actually *decreases* with increasing CR. For small CR (less than 4) the trend is opposite, in agreement with the intuition.

9 C_{pi} and C_{pe} may be related to the wall curvature, allowing their prediction without actual flow calculations. When the wall curvature is small the C_p's depend only on a parameter defining the appropriate cubic and are independent of the other, matching, cubic. This relation holds fairly well up to $C_p = 0.1$. At higher wall curvatures the presence of the other cubic influences significantly the C_p values, which then scale with a different parameter. This latter parameter was determined using a simple analysis.

10 It is conjectured that when C_{pi} and C_{pe} are chosen so that separation at both ends is just avoided, the exit boundary layer thickness is near its minimum.

11 The application of this design to rectangular sections is not straightforward. However, for not too high aspect ratios, one can expect that a comparison with an axisymmetric nozzle

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of an "equivalent" diameter

$$D_{eq} = 2\sqrt{A_1/\pi}$$

will yield a satisfactory estimate of average conditions. The *trends* of the C_p's with CR, L/D_{eq}, and X, if not their values themselves, should be about the same as in the axisymmetric case.

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